

GLaDOS

Geometric Lattice Deduction Over Streams

*a sound, abstain-capable deduction organ — and the evidence that an LLM can be made to **causally** reason through it*

ember + Claude (Opus 4.8) · draft 2026-06-26

Honesty banner. We write “empirically sound on this distribution,” never “certified.” Soundness here is a **property of an output check we run**, not a theorem about the network. The borrowed parts are named as borrowed: the lattice / abstract-interpretation narrowing + abstain + on-policy targets are the Lattice Deduction Transformer (LDT, arXiv:2605.08605); the coupling primitives (zero-init gate, structured readback, freeze-the-reasoner) are TransNAR (arXiv:2406.09308); the process-reward recipes are RLTT (arXiv:2602.10520) and LSRL (Findings EMNLP 2025). What is **ours** is the combination and the discipline: a per-problem-compiled, output-checked deductor woven into a pretrained LLM’s residual stream and verifier-trained, with adversarial causal controls run on every result. Negatives in this paper are results, not omissions.

Abstract. We ask whether a learned, differentiable, output-checked deduction module can be woven into a pretrained language model so that the model’s own generation reasons **through** it — and we hold ourselves to the standard that broke the closest prior work (SATNet): a sound module can still be **bypassed**, so only causal controls (shuffle / permute / corrupt the module’s state) prove genuine deduction. We build an exact CSP harness as ground truth, a bank of three deduction organs (narrow / chain / energy), and a staged training recipe. Our findings, with the negatives: coupling a frozen LLM to a checked deductor gains calibrated abstention it otherwise lacks (88.8% vs 1.4%); RLVR fixes **calibration**, not **capacity**; 7B behaves like 1B on the coupling (capacity is not the wall); **cold co-training fails** (the LM shortcuts the text and ignores the organ) — but a staged recipe (organ→excellence → freeze → establish readout) makes a **learned, general** organ causally load-bearing across seven reasoning types (woven 98.8% vs base 48% vs text-LoRA 25%; corrupt→2.7% **with the full problem text present**). We position this against TransNAR, Coconut, SATNet, and the process-reward line, and lay out the single most decisive open experiment: does ordinary next-token continued-pretraining exploit the organ (a general architecture) or only targeted RL (a trick)?

1 The bet, in one paragraph

Language models confabulate because nothing in their forward pass is allowed to say “I cannot conclude that.” The bet of GLaDOS is that you can give a transformer a **reasoning channel that can abstain** by welding a small, learned deduction module into its residual stream — and that the module can stay **reliable** not because it is a verifier, but because we **check its output**. The check is the whole trick. A learned module is free to be loose, general, even emergent; if a cheap exact check sits at the boundary, the only outputs that survive are the correct ones, and the rest become honest abstentions. We call the module an *organ* and the assembled system GLaDOS.

The reason to want this **woven**, rather than as an external SAT/SMT tool the LLM calls in text, is bandwidth. A text round-trip to a solver is non-differentiable and lossy: the LLM never sees the solver’s internal certified state, gets no gradient on “did I state the problem right,” and inherits none of the solver’s $\perp$ (don’t-know) channel. Weaving keeps the interface differentiable end-to-end and lets the LLM’s **own** head generate from a hidden state already saturated by the deduction.

1.1 The spine

The organ’s state a_t is a point in a per-position *powerset lattice* — for each variable, the set of candidate values still alive. One step of deduction proposes, then commits only by **narrowing**:

$$a_{t+1} = a_t \cap \Pi(f_\theta(a_t, p, m))$$

Here f_θ is a **learned** cell (it proposes, loosely), Π projects the proposal into the lattice, and the meet $\cap$ guarantees $a_{t+1} \subseteq a_t$: the step can only **remove** candidates, never add one. The bilinear/learned part **proposes**; the lattice meet is the sole **authority** on elimination. That single algebraic fact — monotone narrowing — is what makes the recurrence sound-by-construction **early in training** (it is the training-wheels inductive bias), and it is what an exact verifier at the output can later relax: the endpoint is a general, emergent reasoner that stays sound because the **boundary** checks it, not because every internal step is provably correct.

1.2 The one-sentence architecture

At layer k of the host LLM, a learned map α **compiles** the hidden state into a *typed factor-graph program* — variables, domains, and factors, **not** a pre-solved answer; the **organ narrows** that program recurrently, **between** host layers (no backprop through repeated host passes); a learned map γ **writes** the certified state **densely and structurally** back into the residual stream through a **zero-init gate**; the host’s **LM head generates** the answer from the saturated hidden state; an **exact verifier** supplies the reward.

Two phrasings of the same discipline run through the whole paper. First: **soundness is a permission, not a prescription**. The output check buys correctness-given-the-organ; it lets us make the organ loose, learned, and general instead of a hand-built verifier. Second — and this is the lesson that cost us the most — **a sound organ can still be bypassed**. The check guarantees the answer is right **if the organ produced it**; it says nothing about whether the organ, versus the prompt text or a LoRA, actually produced it. Only adversarial causal controls (shuffle / permute / corrupt the organ’s state and watch accuracy collapse) prove the deduction is load-bearing. We run them on every result, and they are the real contribution.

2 The harness: an exact world to be sound about

Before any neural claim, we need ground truth — a generator of constraint problems where the **exactly correct** deduction is computable, so “the organ eliminated a true candidate” is a measurable event, not a matter of opinion. The harness (`clair/csp.py`, `clair/levels.py`) is that world, and it is deliberately unglamorous: its job is to make our soundness claims falsifiable.

2.1 Exact deduction, by construction

For a constraint satisfaction problem p , the harness computes the **exact deduction closure** ded_p : the per-variable set of values that appear in **at least one** full solution. This is the tightest sound narrowing possible — anything tighter would kill a value that some solution needs. Two facts about it are load-bearing and are grounded in the LDT line’s Lean formalization (arXiv:2605.08605):

- **Soundness is free if you check; completeness is the hard part.** A narrowing that only removes values no solution uses is sound by definition. The interesting, **learnable** quantity is *completeness*: how close to ded_p the organ gets — i.e. whether it reaches a checkable answer or honestly abstains. Completeness, not soundness, is the research variable.
- **Per-cell arc-consistency equals exact deduction only at the right level.** Cheap local propagation (per-cell AC) recovers ded_p on binary-structured problems but **not** on problems with higher-arity structure. The harness exposes this as a lattice of *abstraction levels*: level-0 is per-cell/unary, level-1 is pairwise, level-2 carries genuine factor (3-way+) structure.

2.2 The general factor lattice, and the affine wall

The original LDT formulation is binary (pairwise exclusion). Our harness generalizes it to a *factor lattice of any arity*: a factor is a tuple of variables plus an allowed-tuples table, and narrowing propagates through factors, not just edges. This matters because of one recurring obstacle we name the *affine wall*.

The affine wall = width. Three-way / modular-arithmetic / XOR-like constraints (the “affine” structure, e.g. $x + y + z \equiv 0$) cannot be captured by **any** amount of pairwise propagation: per-cell AC is provably stuck. They require a **grade-3 / ternary** primitive in the per-step computation. In the harness, the factor lattice “breaks XOR’s affine wall at level-2” — the factor structure is exactly what supplies the missing arity. The wall is therefore a **width** (per-step expressivity) limit, not a **depth** one; see §4 and the macro-deduction result for why depth alone cannot manufacture it.

2.3 The polymorphism diagnostic

Constraint tractability has a classical algebraic signature: a CSP is tractable iff its constraint language admits certain **polymorphisms** (closure operations). The harness ships a polymorphism diagnostic so that, for any generated instance class, we can tell **in advance** whether per-cell propagation should suffice or whether factor-level structure is required. This is what lets us aim experiments at the regime a method targets — a discipline lesson learned the hard way (don’t conclude a method failed from a mis-aimed run).

2.4 The verifier ladder

On top of the CSP core sits a ladder of exactly-verifiable tasks that double as the training curriculum and the generality test (`curriculum.py`, `fol.py`, `smt.py`, `schedule.py`): a forward-chainer giving entail / contradict / unknown labels, a **z3** SMT oracle, a multi-task scheduler, and Bedrock-rendered natural-language surface forms so the same logical problem appears in many phrasings. The diversity **is** the generality measurement: hold out a phrasing or a problem size, and OOD soundness becomes observable.

3 The woven organ

The architecture is one forward pass up the host’s layer stack, with the organ spliced in as a recurrent side-loop. The temptation — and the trap — is to make α powerful enough to **solve** the problem and γ wide enough to **dump** an answer. A GPT-5.5 review of an early “dense weave” caught exactly this: unstructured dense surgery into the residual stream produces a **new** bypass, where the LM encodes the answer through a lattice-shaped tensor without ever learning stable deduction. The corrections below are not cosmetic; each closes a bypass.

3.1 α compiles a program, it does not solve

α maps the host’s hidden state to a **typed factor-graph program**: the variables, their domains, and the initial factors/evidence — and **nothing about the answer**. If α were allowed to emit candidate-survival logits, “the answer posterior shaped like a lattice” would be a bypass: the organ would have nothing left to do. So α states the problem; the organ narrows it. This compile step is also the part with **no** prior art — TransNAR’s reasoner ingests a clean external graph the LLM never has to produce (§5). α is the unprecedented half, and the staged recipe pretrains it on its own.

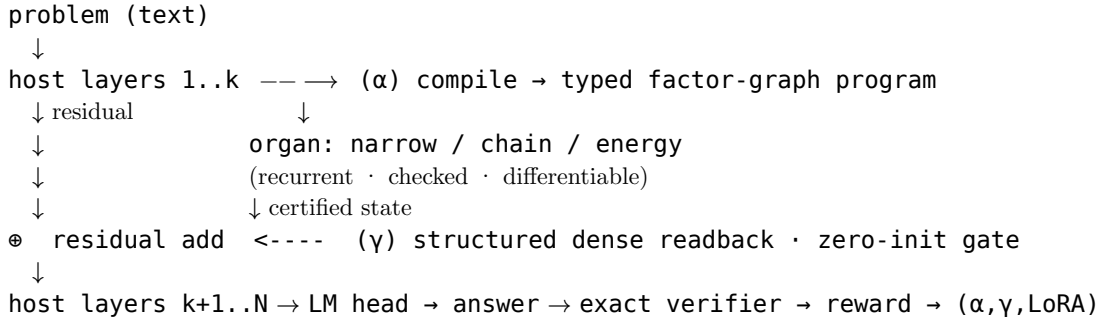
3.2 The organ narrows between layers — the host is not the recurrence

The organ runs its narrowing recurrence **between** two host layers, not by re-running the host. We **do not** backprop through repeated full host passes; old lattice states are stop-grad / truncated, and we supervise every organ step directly. The host drifts the state coarsely across blocks while the organ refines it finely within the loop — a two-scale split that matches the observed dynamics of recurrent-depth transformers (arXiv:2509.23314). The natural halting signal is **monotone**: candidate-set cardinality only decreases, so a cardinality-plateau (with a two-step confirm) is a cleaner early-exit than the norm-threshold halts that misfire on generic recurrent spirals.

3.3 γ reads back densely and structurally, through a zero-init gate

γ writes the certified lattice state back into the residual stream. Two design choices, both corroborated by TransNAR (arXiv:2406.09308):

- **Structured, not flattened.** Flattening the lattice to a fixed $[N, K]$ tensor is anti-equivariant and tied to a maximum problem size — it wrecks size generalization. γ instead uses equivariant adapters over the explicit cell-candidate **and** factor-tuple tensors. TransNAR found the same: reading node **and** edge structure and combining by concat+linear beats every reductive pooling (sum, MLP, Gram-Schmidt).
- **Zero-init gate.** The readback is gated with the gate initialized **closed** (Flamingo / LLaMA-Adapter style), so step-0 of the hybrid is the host **exactly** — no damage to pretrained knowledge — and the model learns to open the gate only as the organ proves useful.



Listing 1: The weave. α compiles (does not solve); the organ recurs between layers (the host is not the recurrence); γ reads back structurally through a closed-init gate; the host’s own LM head generates; an exact verifier closes the loop. Gradient reaches α , γ , and the host LoRA — the organ itself is frozen during weaving (see the staged recipe).

3.4 The organ bank: a shared protocol, three inference shapes

There is not one organ but a bank, sharing a **protocol** (compile → recurse → check) rather than a coupling — each has its own state schema, halting rule, and verifier:

- **narrow** (proposer.py) — monotone candidate rule-out; sound, abstains. Best for satisfaction.
- **chain** (chain_organ.py) — **grows** derived facts instead of narrowing candidates (the dual); best for entailment/derivation. Exhibits a sharp phase transition: sound **or** deep, rarely both at once.
- **energy** (energy_organ.py) — soft constraint satisfaction as an equilibrium; uniquely does **optimization** (85.5% exact min-cost), with 6% declared infeasible at the affine wall.

No single organ dominates; a portfolio plus verifier-selection is complementary. The LM router alone is not trusted to choose — the verifier arbitrates.

3.5 Geometric algebra, placed precisely

Geometric algebra was the project’s orphaned opening thread, and as a **generic channel mixer** it is a clean null (it ties SwiGLU/MLP in language and vision). Its real home is the factor deductor, via a structural identity: **geometric-product grades correspond to constraint arities**. Grade-0 (inner product) $\leftrightarrow$ unary; grade-2 (the wedge $u \wedge v$, antisymmetric, “these two cannot coexist”) $\leftrightarrow$ binary exclusion; grade-3 (the trivector $u \wedge v \wedge w$) $\leftrightarrow$ the ternary/affine primitive that breaks the level-0 wall. The right abstraction is the **un-truncated fold** — one wedge spanning all arities up to a capacity K — which matches hardcoded grades in-distribution and generalizes best zero-shot. Crucially, GA is a **completeness / representation** bias, **not** a soundness fix: soundness comes only from the output check and typed symbolic kernels (e.g. exact modular arithmetic), never from the algebra (arXiv:2601.06793).

3.6 The staged recipe — the central lesson

Stage 1 — organs to excellence, separately. Train each deductor hard across many tasks until it is a general, robust, near-oracle **standalone** reasoner (no LM). Cheap: pure organ.

Stage 2 — pretrain each interface piece on its own sub-task. γ on oracle-lattice $\rightarrow$ answer readout (proven; see §4); α on text $\rightarrow$ program extraction (supervised). Each translator must be competent **before** it has to cooperate — and each gets its own isolated grounding test, never inferred from end-to-end numbers (the SATNet lesson).

Stage 3 — assemble, then consolidate. Plug the pretrained pieces together, **freeze the organ**, and only then run the long, weird training where the LM learns to route its reasoning through the organ — closer to re-running real pretraining (Dolma-with-the-organ) so organ-use becomes **native**, not bolted-on.

Figure 1: The staged recipe. The central finding (§4) is that you must **not** cold-co-train the LM and the organ — two alien substrates, both bad at init, cannot bootstrap together and the LM settles into a text-shortcut. Decouple: make the organ **good** (pretrained, frozen) and **necessary** (the task cannot be shortcut). The organ becomes load-bearing only when both hold.

3.7 The organ as its own process reward

Because the organ narrows **monotonically** and **checks** each step, it is its own exact per-step grader: the process-reward “progress” signal is the candidate-set cardinality drop at step t , and the “quality” signal is the soundness of that drop (guaranteed by the check). This is exactly the dense per-step supervision that the process-reward line (RLTT/LSRL, §5/§6) had to approximate with a hackable LLM judge — and here it is free, exact, and unhackable.

4 Training: the staged, oracle-supervised recipe

The architecture (§3) says **what** the weave is; this section says **how** it is taught. The recipe is **staged, curriculum-driven, and oracle-supervised** — and that last word matters: it is **not** classic semisupervised learning. Every label is **free and exact**. Witness-first generation hands us a known solution, and the exact per-cell deductor (`clair.csp.exact_dedP`) hands us the exact narrowing targets, the answers, and the verifiable rewards. There are **no human labels and no noisy pseudo-labels** — the “curriculum” is purely the **staging order**, the readout-forcing schedule, and the domain / composition / N mix. The entry point is `clair/organ/train.py` (organ for Stage 1, **weave** for Stages 2–3).

4.1 Stage 1 — pretrain the organ, standalone

The deductor learns to **narrow soundly**, with the base LLM not yet involved. The loss is **dominate-ded_p**: a soundness-asymmetric supervised loss against the exact per-cell transition. Eliminating a value that some solution actually uses (an **unsound** drop) is penalized heavily ($w_{\text{pos}} \approx 6$); keeping an extra candidate (**loose but sound**) is penalized lightly ($w_{\text{neg}} \approx 0.5$). The step is monotone — survival only ever decreases (the meet) — so the asymmetry trains the organ toward the sound side of the boundary. Training is **on-policy**: each batch is the previous step’s narrowed lattice rolled forward by the organ, with targets re-computed by the exact deductor every pass (the **given** set is recomputed after each meet).

The data is the **broad multi-domain corpus**: all beast generators (CSP families $\cdot$ xor/GF2 $\cdot$ modular $\cdot$ graph $\cdot$ permutation $\cdot$ type $\cdot$ FOL $\cdot$ optimize), plus **union relations** (named families **and** random allowed-tuple tables, added **additively** — this is the fix for OOD-novel-relation soundness), wide N , and **30% composed problems** (multi-skill). The sweep recipe is **1.5M parameters, $R = 12$ recurrent steps** (with an $R = 8$ floor) and the 30% composition mixup. Composition **recall** is nearly free (the meet composes); composition **coverage** buys **soundness** on novel compositions — a need that **grows with scale**. The output is a frozen, general, sound narrower checkpoint (`runs/general_organ_full.pt`).

4.2 Stage 2 — freeze the organ, weave the live readout

Now the LLM learns to **generate through** the frozen organ. The Stage-1 organ is **frozen** (a slow-thaw / EMA variant is studied; default is freeze), the OLMo base is **frozen**, and the only trained parameters are **LoRA + latent- α + γ** :

- **latent- α** is a dense projection from the OLMo hidden state to the organ’s input tensors. It is **fully latent** — there is **no symbolic extraction step**. The hidden state compiles directly to the typed factor-graph program (variables, domains, factors), and never to the answer.
- γ is the structured readback, with its gate **zero-initialized** (closed) so step-0 of the hybrid is the host exactly.

The forward pass is: OLMo hidden $\rightarrow$ latent- α $\rightarrow$ the frozen organ narrows (under `no_grad`, so we **never** backprop through repeated OLMo passes) $\rightarrow \gamma$ injects the narrowed lattice into the residual $\rightarrow$ the LM head generates the answer. The loss is plain **LM cross-entropy on the answer span only**. α never gets a direct target; it learns purely from the gradient flowing **through the organ’s downstream success** — the LM does better when α has compiled a usable problem, and that signal is what teaches the compile.

The curriculum here is **readout-forcing**. Training starts in **cells-mode** — no facts in the prompt, so the organ is the **only** route to the answer — which forces the γ -decode to form **before** the LM can settle into a text-shortcut basin. Full-text problems are then mixed back in. The output is a woven model in which the LM **causally** wields the organ: corrupt the lattice and generation collapses.

4.3 Stage 3 — RLVR and consolidation

Two moves sharpen and stabilize the woven model:

- **RLVR** (GRPO; verifiable reward = the exact checker) with the **organ-as-process-reward** (LSRL-style: per-step candidate-set cardinality drop), mixed **0.7 outcome / 0.3 process** so the per-step signal shapes rather than dominates. This sharpens **calibration**, not capacity (§4); we report pass@1 **and** pass@k with ProRL and random-reward controls, on OLMo as the honest base (§6).
- **Consolidation**: continued-pretrain on **Dolma $\oplus$ a reasoning corpus** (general $\oplus$ organ-shaped, tuned ratio) with the organ in place, so organ-use folds into general behavior **without** catastrophic forgetting — the preserve-and-generalize endpoint.

4.4 The honest correction: readout-isolation vs live latent- α

Scope correction. The woven numbers reported in §4 — woven 98.8%, corrupt $\rightarrow$ 2.7% with full text present — are **readout-isolation** results: the organ is handed the **true** narrowed lattice (the ground-truth CSP, compiled outside the model), and what is proven is that the LM **reads back and generates through** a certified state it cannot get from the prompt. That readout is real and causally controlled, and it is the hard half TransNAR also enjoys (a clean external graph it never has to produce). The **real** woven model closes the remaining gap with **live latent- α** : α compiles the organ’s input **from the OLMo hidden state**, with **no symbolic extraction and no ground-truth CSP**. That model is being wired now, post-codex-preflight. **Live latent- α is the critical path and the honest open piece** — readout is proven; α (latent compile) is the hardest part, and a wrong compile means a confidently wrong organ.

4.5 The base matrix — one organ, many hosts

The pretrained organ is **base-agnostic**: Stage 1 is done once, and the woven readout (Stages 2–3) is a weights-only bolt-on, so each base is a cheap, independent paper datapoint. The set now spans **100M $\rightarrow$ 32B across six model families**, including an SSM-hybrid, to show the technique uplifts across scale **and** architecture rather than on one lucky host.

base	tier / family	why it is in the matrix
Pythia-160M / SmolLM2-135M	100M · Pythia/SmolLM	ships all training checkpoints = interp-gold; the organ-carries-the-reasoning extreme (host too small to reason alone)
OLMo-2-1B	1B · OLMo	cheap full-recipe datapoint, fully-open base; low end of the scale curve
SmolLM3-3B	3B · SmolLM	fully-open small-modern model; a clean small datapoint between 1B and 7B
OLMo-3-7B (PRIMARY)	7B · OLMo	interp anchor, full causal controls, the honest fully-open mechanism — run #1, never skipped
OLMo-3-32B	32B · OLMo	does the uplift hold / grow with base size (QLoRA @ 4-bit)
Gemma-4	mid · Gemma	portability + uplift on an already-strong, code-trained model
Qwen-3.6	mid · Qwen	portability across a second strong family
Nemotron-H-8B	8B hybrid · Nemotron	the architecture-diversity datapoint — Mamba-2 + self-attention 1. MLP hybrid; does the residual-stream coupling survive when the splice point is an SSM block, not attention? Also fully open (weights + data + recipe), so interp-friendly like OLMo

Table 1: The base matrix. 2.5 orders of magnitude (100M→32B) over six families (OLMo · Gemma · Qwen · Nemotron-H · SmolLM · Pythia), including an SSM-hybrid. Compute: woven = frozen base

1. LoRA + α + γ ; 7B/8B fit one L40S, 32B fits one A100-80 (or L40S @ 4-bit), the $\leq 3B$ tier runs anywhere. OLMo-3-7B is run #1.

4.6 What each stage proved, and the standing caveats

The evidence ladder (§4, in order): the readout works (corrupt→0, oracle de-risk); a **frozen learned organ** reads as well as the oracle; **cold co-training fails and staging fixes it**; the staged generative model uses the organ across seven rung types (corrupt→2.7% with text present); next-token CPT recruits the organ **with no reward** (architecture, not RLVR trick); and the split-brain control shows the LM wields the organ 99.9% over adversarial text (necessity, not opportunism). The standing caveats stay on the page: the organ’s soundness is **trained**, not by-construction (the certified operators are the floor, the net is guidance — **neural proposes, certified guard narrows**); the model trusts the organ **unconditionally** (split-brain), so **α -correctness is the whole ballgame**; and the **affine wall** is real — single ternary constraints need grade-3 and XOR **systems** need the certified row-space organ, neither beaten by depth alone.

5 Findings — the honest arc

This is the project’s spine read in order, negatives included. The shape of the arc is: the **coupling** works on a frozen LLM; RLVR and scale are **not** the levers people assume; the naive end-to-end version **fails**; and a staged version **works**, validated by causal controls — which is the only thing that distinguishes it from an award-winning artifact that was secretly reading the labels.

5.1 The checked-deductor coupling works (and what RLVR / scale actually buy)

A frozen LLM bolted to a checked deductor gains a capability it structurally lacked: **calibrated abstention**. Deterministic accuracy rises to **88.8% vs a base 1.4%** — the base model essentially cannot say “I can’t conclude that,” and the organ gives it that channel. Three sharpening results follow, each a “not what you’d guess”:

- **RLVR fixes calibration, not capacity.** RL adds +18–20 points of abstain-recall OOD — it tunes **when** to defer — but does not raise the underlying solvable set. (This matches the broader RLVR-elicits-not-expands picture, [arXiv:2504.13837](#) see §6.)
- **7B $\approx$ 1B on the coupling.** Scaling the host from 1B to 7B does not improve the **reading** of the organ. Capacity is not the wall; 7B buys LM competence, not better deduction-use.
- **Grounding fixes extraction robustness, not exactness, and set-valued readout makes degradation graceful and sound** — when the organ is unsure it returns a **set**, and the system degrades by abstaining rather than by confidently erring.

5.2 Cold co-training fails — and staging fixes it

The decisive negative: a **learned** organ co-trained from scratch with the LM is **ignored**. The LM settles into a lazy local optimum — shortcut the answer from prompt text, fall back on an abstain prior — and the causal controls go flat (the organ is decorative). An “airtight” bootstrap run confirmed this even with a necessary task, a perfect organ, and a forced-open gate. This is the same disease as SATNet’s grounding collapse (§5): wherever the loss admits a cheaper non-deductive optimum, it is taken.

The fix is **not** to deny the LM the text. It is to **decouple training**: bootstrap the organ to excellence, **freeze** it, then establish the readout. Two intermediate results pinned the diagnosis:

- **Oracle readout (de-risk, decisive, positive).** Feed the LM the **true** narrowed lattice through structured γ , and its LM head generates the right answer **causally** — corrupt the lattice and accuracy goes to 0 **even with the full problem in the prompt** (the LM ignores the text and reads the lattice), 100% OOD. The “latch gap” was a **bandwidth** problem; structured dense γ solves it.
- **Frozen-learned-organ readout (positive, reconciling).** A **frozen, standalone-trained learned** organ reads through γ essentially as well as the exact oracle. So the co-training failure was **distrust of a moving organ**, not the coupling itself.

5.3 ★ The staged generative GLaDOS works

Putting the recipe together yields the project’s positive deliverable: a **learned, general** organ causally woven into the LM’s **generation** across seven reasoning types (coloring, equality, ordering, arithmetic, all-different, equality-chain, forced-color).

Woven 98.8% · base 48% · text-LoRA 25% in-distribution; OOD-phrasing **96.3%**; cells-with-no-facts **98.7%** vs chance. The decisive control: **corrupt→2.7%** with the **full problem text present** — the LM ignores the text and wields the organ. Per-rung true→corrupt drops of 85–100 points, **including hard propagation chains the base and text-LoRA cannot**

shortcut (woven 100% there). OOD-by-size is limited by the **organ’s own recall**, not the readout. The staged recipe is **demonstrated, not hoped**.

Scope, stated plainly. These woven numbers are **readout-isolation**: the organ is handed the **true** narrowed lattice and what is proven is that the LM reads back and generates **through** a certified state it cannot get from the prompt. The **real** woven model replaces that hand-in with **live latent- α** — α compiling the organ’s input from the OLMo hidden state, no symbolic extraction, no ground-truth CSP — which is being wired now (§7). Readout is proven; live latent- α is the critical path and the honest open piece.

setting	woven	base	text-LoRA	reading
in-distribution (7 types)	98.8%	48%	25%	the organ generates, not the text
OOD phrasing	96.3%	—	—	robustness from the organ, not surface form
cells, no facts given	98.7%	chance	—	vs a chance floor
corrupt, full text present	2.7%	—	—	collapses → the organ is load-bearing
det-acc, frozen-LLM coupling	88.8%	1.4%	—	calibrated abstention the base lacks

Table 2: Headline results. The single most important row is **corrupt, full text present**: accuracy collapses to near-zero even though the problem is fully stated in the prompt — the proof that generation is routing through the organ rather than reading the text.

5.4 The causal-control table — the actual contribution

Every woven result is gated behind adversarial controls (`oracle_readout.py`). A **sound** organ guarantees a **correct** answer if the organ produced it; it does **not** prove the organ produced it. These controls do.

control	what it injects / what it proves	result
true	the real narrowed lattice — the intended path	pass
shuffle	a different problem’s lattice (roll along batch); unchanged accuracy → bypass	collapses
permute	permute candidate/value labels; tests intended semantics, not an alias	collapses
corrupt	flip only the query cell’s survivors to a wrong set; must change the answer	→ 0

Table 3: The causal controls — exactly the ones SATNet did not run. The dense- γ oracle result is the SATNet grounding test **passed**: the LM reads **this** problem’s certified state and ignores the prompt text. The open job is keeping it passed once α and a learned organ replace the oracle.

5.5 A (explicit) vs B (latent): the bitter-lesson path

We tested a “bitter-lesson” variant B, where the organ is fed **only** the LM’s hidden state of the text — no extracted factors, no extraction supervision. In-distribution it narrows 94% of **ded_p** (within 5 points of explicit-from-factors). The honest verdict from the diversity sweep:

- Wide- N training **collapses the OOD soundness failure** (false-elim 8.8%→0.9%); multi-task training **recovers phrasing** robustness (recall 76→88, false-elim 14%→3%). The latent narrower is **provably equivariant** (perm/size error = 0), so any residual leak is in α (text→latent), not the architecture.

- What diversity does **not** close: OOD-by-size **recall** stays 65–74% for B vs explicit’s 93%. B is a sound, phrasing-robust, credible bitter-lesson path; **explicit retains a 20-point OOD-completeness edge** (it gets size-invariant factors for free).

5.6 Macro-deduction: log-depth on the bounded-width side

Checked **macros** give sublinear-depth sound reasoning: $\lceil \log_2 L \rceil$ macro-applications replace L base steps, with an **identical exact ded_p fixpoint, 0 false-elim, verified to $L = 256$** (a $32 \times$ compression at $L = 256$). The compression is **symbolic transition-monoid composition** (graph-squaring, arXiv:2210.10749), not the neural net — which is only a fidelity probe (sound to $k = 8$). Cost is polynomial in the state space $S = k^w$, so it works **only on the bounded-width side: width is the wall; bounded-width depth is a log-depth shortcut**. Geometry gave no edge here (the automata transitions are too simple) — an honest null for GA-helps-macro.

5.7 Blades and the recurring null

Blades (the GA grades) help **affine generalization** specifically, consistent with grades→arities. And the recurring signal across all three organs: the **affine wall (3-way / arithmetic correlation) is the level-0 ceiling** — the place where factor / grade-3 structure must earn its keep. The geometric product as a generic mixer remains a clean null; its value is structural, not magical.

6 Related work and the novelty delta

GLaDOS is a **combination**, not a from-nothing primitive — every ingredient has a precedent, and being honest about which is the whole point of the honesty banner. What no one holds is the **five-way** combination, and what no one else **ran** is the discipline. We position against the four nearest neighbors.

6.1 TransNAR — the coupling cousin (but soft, and it skips the hard half)

TransNAR (arXiv:2406.09308) couples an LLM to a **frozen, pretrained** graph reasoner via zero-init gated cross-attention, trained with **pure supervised next-token, no RL**, for >20% absolute OOD gains. It validates our coupling design almost line-for-line: gate init closed, read node **and** edge structure by concat+linear (pooling loses), reasoner frozen during coupling, and — a fix it handed us — **randomized positional encoding is load-bearing**: without it the hybrid is thresholded by the base LM’s OOD score, a candidate cause of our cold-weave shortcut. Two deltas: TransNAR’s reasoner is **soft** (a CLRS-pretrained net, wrong OOD — its index/search failures are exactly that), where ours is **output-checked → sound**; and its reasoner ingests a clean external graph the LLM never produces. Our α (hidden→typed program) is precisely the input-production step TransNAR sidesteps — the unprecedented half.

6.2 Coconut — our mechanism, emergent and unchecked

Coconut (arXiv:2412.06769) makes a **latent** “continuous thought” loop load-bearing under ordinary LM loss — **but only with a multi-stage curriculum** that progressively replaces text CoT with latent thoughts. **Remove the curriculum and the module is ignored** (GSM8K 34.1%→14.4%; ProntoQA 99.8%→52.4%). This is the cleanest external analogue of our cold-co-train failure **and** of our staging fix: LM loss **can** exploit a reasoning module, but a naive single-stage objective collapses to the shortcut. The delta: Coconut’s latent loop is a **free-form superposition-elimination** with no check — exactly our mechanism, but **emergent and unverified**. Ours is checked, so its degradation is **abstention**, not silent error.

6.3 SATNet — the cautionary tale, and the discipline lesson

SATNet (arXiv:1905.12149) was a differentiable MAXSAT layer that appeared to learn logic end-to-end, including “visual Sudoku” with no digit labels. The grounding critique (arXiv:2312.11522)

showed the “learned logic” was **label leakage**: the loss term on input bits backpropagated the true digit labels straight into the CNN through an unmasked side channel. With that channel masked, visual Sudoku collapses to **0.0% $\pm$ 0.0%** over 10 seeds; a plain FC net **beat** the SATNet model (72.1%). The reasoning-shortcut theorem (arXiv:2305.19951) then proved the general result: **reasoning shortcuts are optima of the loss** — a NeSy model can attain perfect accuracy with unintended concept semantics, right for the wrong reasons.

The discipline lesson. SATNet is the field’s most reproducible result about systems like ours, and it is a **negative**: a **sound** organ can still be **bypassed**, and end-to-end differentiability **hides** it. The soundness check is therefore **necessary but not sufficient** — it buys correctness-given-the-organ, never load-bearingness. Our edge over the whole field is not the organ; it is the **causal controls (corrupt / shuffle / permute)** — exactly the controls SATNet did not run — gating every claim, plus isolating α with its own grounding test before co-training, and reporting over seeds. The dense- γ oracle result is the SATNet test **passed**.

6.4 RLTT / LSRL — process reward, and the grader they wanted is our organ

The process-reward line fixes a credit-assignment bug we will hit: outcome-only GRPO credits **only the terminal state** and so **cannot reward an organ’s internal steps** (RLTT, arXiv:2602.10520 LSRL, Findings EMNLP 2025). RLTT broadcasts one outcome reward across loop steps parameter-free; LSRL builds a genuine per-step reward by decoding and grading each latent state — **with a hackable GPT-nano judge**, and it explicitly asks future work for “a symbol-aware PRM verifying each step.” **That is the GLaDOS organ.** Process-reward = candidate-set cardinality drop; step-quality = soundness of the narrowing. Exact, unhackable, free. The novelty delta in one line: **TransNAR is soft, RLTT has no verifier, LSRL’s judge is hackable — ours is exact.**

6.5 The five-way combination nobody holds

Surveying the differentiable-symbolic-module cell (NeSy $\times$ LLM survey arXiv:2508.13678 nearest named cousins DiLA and the VSA/HRR readout arXiv:2502.01657 reward-side RLSF arXiv:2405.16661), no single system holds all five of:

property	who is missing it
(i) woven in the forward pass	DiLA, RLSF, tool-LLMs are external
(ii) differentiable, dense structured readback	tool / abductive / post-hoc-verifier methods are not
(iii) output-checked $\rightarrow$ sound	SATNet, TransNAR, AutoCoNN, the VSA readout are soft
(iv) a per-problem compiled typed program	DeepProbLog/LTN/SATNet use fixed/learned-but-fixed rules
(v) verifier / RLVR-trained	only RLSF + the tool-verifier line — and those are not woven

Table 4: The honest novelty claim: **first to weave a checked, per-problem-compiled deductor into a pretrained LLM and verifier-train it.** The novelty is narrow and integrative — a combination plus the soundness twist, not a new primitive. The defensible edge is the discipline.

7 Toward real training, and the decisive open experiment

The toy is built and it works — but it is profoundly **undertrained** (1B host, synthetic CSPs). The honest next step is real RLVR on standardized verifiable reasoning, with the organ as a **tested** component, on a base model where a gain **means** something.

7.1 The RLVR pivot

Two literature findings reshape **how** we test the organ before **whether** it helps:

- **Use OLMo as the base.** “Spurious Rewards” (arXiv:2506.10947) shows random/wrong rewards recover most RLVR gain on Qwen-Math but **not** on Llama/OLMo — Qwen’s RLVR partly measures pretraining. On OLMo a gain is real signal. We standardize on **OLMo-3-Base-7B** (Base, not Instruct: RL headroom + an honest control; our 7B $\approx$ 1B coupling finding means 7B buys LM competence, not better reading). A random-reward and incorrect-label **dummy arm** is mandatory — on OLMo it should do nothing; if the organ-augmented model suddenly gains under random reward, that exposes an amplifiable shortcut.
- **Report pass@k, not just pass@1, with a ProRL-style control.** RLVR mostly **elicits** rather than **expands** (arXiv:2504.13837): it raises pass@1 and can **lower** pass@k. The capability question is whether the organ behaves like **distillation** (lifts pass@k, expands the boundary) or like plain RLVR (sharpens the prior). A prolonged, KL-controlled, task-diverse RLVR arm (ProRL) is needed so we do not credit the organ for beating a deliberately weak baseline.

The substrate is de-risked: the base TRL-GRPO loop (`rlvr_pipeline.py`) **runs and learns** on one L40S (reward EMA 0.20 $\rightarrow$ 0.72 over 250 steps on `chain_sum`), with `use_vllm=False` for an organ-compatible dynamic-depth rollout, Dr.GRPO, $\beta = 0$, LoRA all-linear. The organ slots in by swapping the policy model; reward, dataset, and config are unchanged. Fuel is reasoning-gym (arXiv:2505.24760) plus ZebraLogic (arXiv:2502.01100) / SATBench for CSP transfer and a MATH slice for cross-domain. The premise is vindicated by Logic-RL (arXiv:2502.14768): a 7B trained **only** on procedural Knights-and-Knives transferred to AIME/MATH.

7.2 The organ as process reward

Because outcome-only GRPO **falsely nulls** a looped organ (it never credits internal steps), a process/trajectory-reward arm is not optional — it is what makes a GRPO null **interpretable**. And the organ **is** the grader: per-step progress = candidate-set cardinality drop, per-step quality = soundness of the narrowing. We adopt both published arms (RLTT broadcast, parameter-free; LSRL exact-graded), honoring the depth floor (process supervision is null below $r = 8$ recurrent steps) and keeping the outcome reward primary (0.7 outcome / 0.3 process) so the per-step signal shapes rather than dominates.

7.3 The decisive open experiment: architecture or trick?

The single most decisive experiment. The organ supplies **correctness**; next-token CPT optimizes **likelihood**. On generic prose the two diverge, so the gradient that would route generation through the organ may be absent (the cold-weave failure). **Does ordinary next-token continued-pretraining exploit the organ — making it a general architecture — or does only targeted RL open the gate — making it a trick?** No published work tests next-token CPT on a **checked deductor**.

The design is a **2 $\times$ 3 grid**: {next-token CPT, process-RLVR} $\times$ {reasoning-dense + non-shortcuttable, reasoning-dense + shortcuttable, general prose}. Starting from a checkpoint M $\star$ where the organ is already causally load-bearing (staged recipe), continue-train with **ordinary next-token loss only** and track the **exploitation gap** $G(t) = \text{loss}_{\text{off}} - \text{loss}_{\text{on}}$ — next-token loss with the organ zeroed vs on, over CPT steps. **Exploited** = $G(t)$ grows monotonically and concentrates on deduction-determined (low-entropy, verifier-checkable) tokens, the γ gate opens from 0, the corrupt $\rightarrow$ 0 penalty **rises**, and OOD verifiable accuracy improves **with no verifier reward in the loop**. **Null (RL-only trick)** = $G(t)$ flat/shrinking and the gate stays shut while the process-RLVR arm on the same data **does** open it. The literature is genuinely split here: the get-go position (arXiv:2502.19402, which proposes our exact reasoner+memory architecture and bets on RL) and SFT-memorizes/RL-generalizes (arXiv:2501.17161) predict the trick; procedural-knowledge (arXiv:2411.12580),

front-loading (arXiv:2510.03264), and distillation (arXiv:2504.13837 Fig. 7) predict the architecture — **given the right (reasoning-dense, non-shortcuttable) data**. The 2×3 grid adjudicates it on our own organ.

7.4 Future architectures

Two host shapes marry the organ better than a dense autoregressive stack:

- **DiffusionGemma — organ ↔ denoising.** A diffusion/denoising LM already runs an **iterative refinement** over the whole sequence; the organ’s monotone narrowing is a natural per-step constraint **projection** inside the denoising loop (narrow the candidate set, then denoise toward it). The organ’s recurrence and the model’s iteration become **one** loop rather than two nested ones — and the monotone exit signal gives the denoiser a principled, sound stopping criterion.
- **MoE — organ as a routed expert.** In a mixture-of-experts host, the organ is one expert the router can dispatch to on reasoning-dense tokens. This makes “use the organ” a **learned routing decision** trained by the same load-balancing machinery, and it bounds the organ’s cost to the tokens that need it — a clean answer to “when should the model reason through the organ” that the router learns rather than the gate.

The consolidation endpoint remains the same in all cases: re-run OLMo’s own open recipe (Dolma continued-pretrain + instruction-tune + RL-Zero) **with the organ in place**, so organ-use folds into general behavior rather than staying a bolted-on skill. Whether **ordinary** training does that is the architecture-vs-trick question above — and it is the experiment that decides what GLaDOS actually is.

Lineage / references (arXiv ids inline above). LDT 2605.08605 · CliffordNet 2601.06793 · automata-shortcuts 2210.10749 · TransNAR 2406.09308 · Coconut 2412.06769 · SATNet 1905.12149 · SATNet grounding critique 2312.11522 · reasoning shortcuts 2305.19951 · RLTT 2602.10520 · LSRL (Findings EMNLP 2025) · looped transformers 2502.17416 · two-scale recurrence 2509.23314 · RLVR-elicits 2504.13837 · spurious rewards 2506.10947 · get-go RPT 2502.19402 · SFT-memorizes-RL-generalizes 2501.17161 · emergent-under-CPT 2506.00288 · procedural-knowledge 2411.12580 · front-loading reasoning 2510.03264 · Logic-RL 2502.14768 · reasoning-gym 2505.24760 · ZebraLogic 2502.01100 · DiLA (OpenReview uh3ZO2izyr) · VSA/HRR readout 2502.01657 · RLSF 2405.16661 · NeSy×LLM survey 2508.13678. Full notes + URLs in [clairnets/notes/](#).